ANALYZING PATTERNS AND ETHICAL IMPLICATIONS OF WORKFORCE LAYOFFS IN 2022

A People Analytics Perspective

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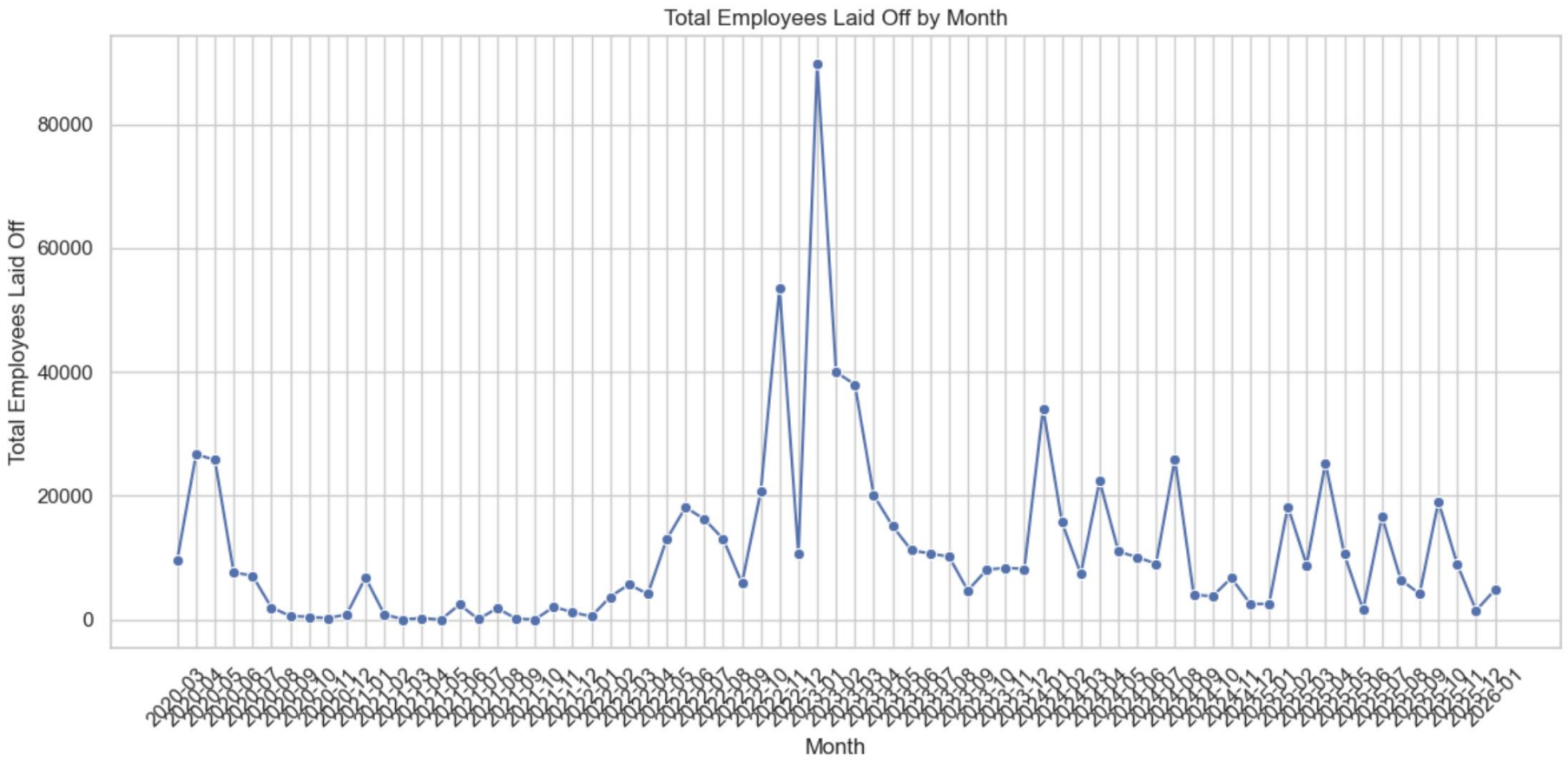
WHY THIS TOPIC MTTERS?

- Layoffs accelerated sharplyin 2022
- Public narratives focus onheadlines and totals
- Human impact is oftenmissing from the data
- People analytics can addcontext and responsibility

DATA SOURCE & SCOPE

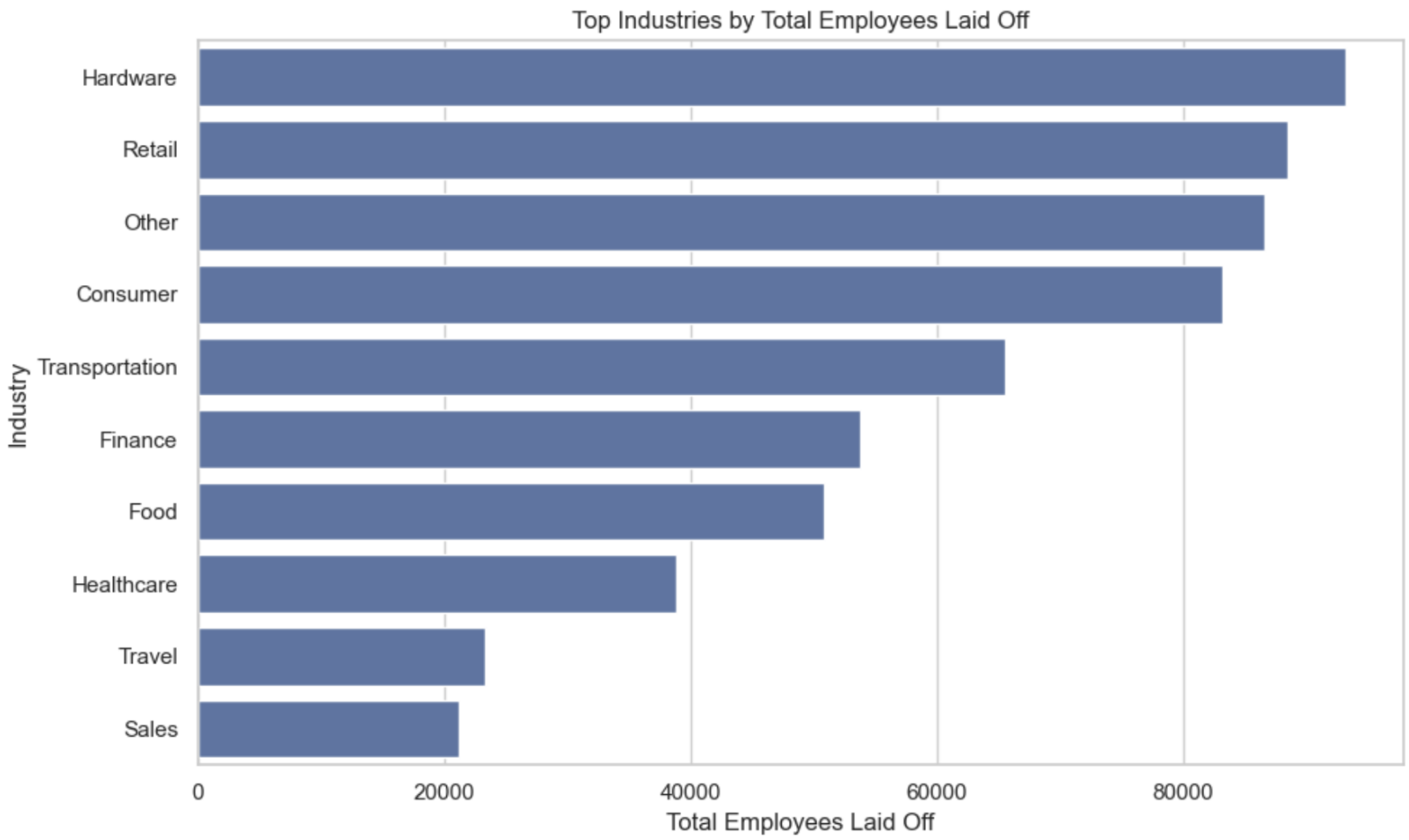
- Publicly reported layoff data (Layoffs.fyi)
- Company, industry, date, size, and % workforce
- Covers reported events, not the full labor market
- Visibility ≠ completeness

TEMPORAL LAYOFF TRENDS



- Layoffs occurred in sharp waves, not steadily
- Peak month exceeded 90,000 employees laid off
- Many months remained below 10,000
- Indicates reactive, shock-driven decisions

INDUSTRY CONCENTRATION



- Layoffs concentrated in a few industries
- Hardware, Retail, Consumer led by total impact
- Industry totals mask variation in event size
- Frequency and magnitude tell different stories

FREQUENCY VS MAGNITUDE

	industry	total_employees_laid_off	layoff_events	median_layoff_size
0	Hardware	93207.0	75	300.0
1	Retail	88411.0	341	100.0
2	Other	86635.0	290	111.0
3	Consumer	83160.0	275	112.5
4	Transportation	65480.0	264	129.0
5	Finance	53824.0	505	71.5
6	Food	50811.0	244	124.0
7	Healthcare	38904.0	326	81.0
8	Travel	23370.0	107	137.0
9	Sales	21193.0	71	70.0

- I'm showing the table here so you can see how industries compare across both the number of layoff events and the typical size of those layoffs
- Finance: many events, smaller median layoffs
- Hardware: fewer events, much larger median size
- Median layoff size varies widely by industry
- Counts alone are misleading

LAYOFF SEVERITY INDEX

	company	industry	employees_laid_off	percent_workforce_laid_off	layoff_severity_index
219	Intel	Hardware	22000.0	0.20	0.760000
534	Intel	Hardware	15000.0	0.15	0.522242
700	Tesla	Transportation	14000.0	0.10	0.475420
58	Amazon	Retail	14000.0	0.01	0.448420
2169	Google	Consumer	12000.0	0.06	0.399775
2653	Meta	Consumer	11000.0	0.13	0.388952
3605	Katerra	Construction	2434.0	1.00	0.377361
1701	Flink	Food	8000.0	0.40	0.374485
2202	Microsoft	Other	10000.0	0.05	0.333130
3211	Butler Hospitality	Food	1000.0	1.00	0.331727

- Combines layoff size and % workforce impacted
- Accounts for proportional employee impact
- Highlights risk beyond headline numbers
- Differentiates large firms vs. smaller organizations

SEVERITY EXAMPLE

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- Intel: 22,000 employees (~20% workforce)
- Amazon: 14,000 employees (~1% workforce)
- Katerra: 100% workforce laid off
- Size does not equal severity

COMPANY STAGE RISK

	stage	layoff_events	total_employees_laid_off	average_percent_laid_off
0	Post-IPO	992	483456.0	0.168143
1	Unknown	718	76457.0	0.381300
2	Acquired	381	70341.0	0.332238
3	Series B	476	30572.0	0.355281
4	Series C	437	27412.0	0.253273
5	Series D	350	27054.0	0.201291
6	Series E	197	24548.0	0.182687
7	Series F	116	15151.0	0.149722
8	Private Equity	76	12676.0	0.182826
9	Series A	268	8932.0	0.476136
10	Series H	36	8497.0	0.151200
11	Subsidiary	29	7538.0	0.280625
12	Series G	27	5039.0	0.152500
13	Series J	11	4700.0	0.134000
14	Series I	9	3155.0	0.087143
15	Seed	140	2531.0	0.816909

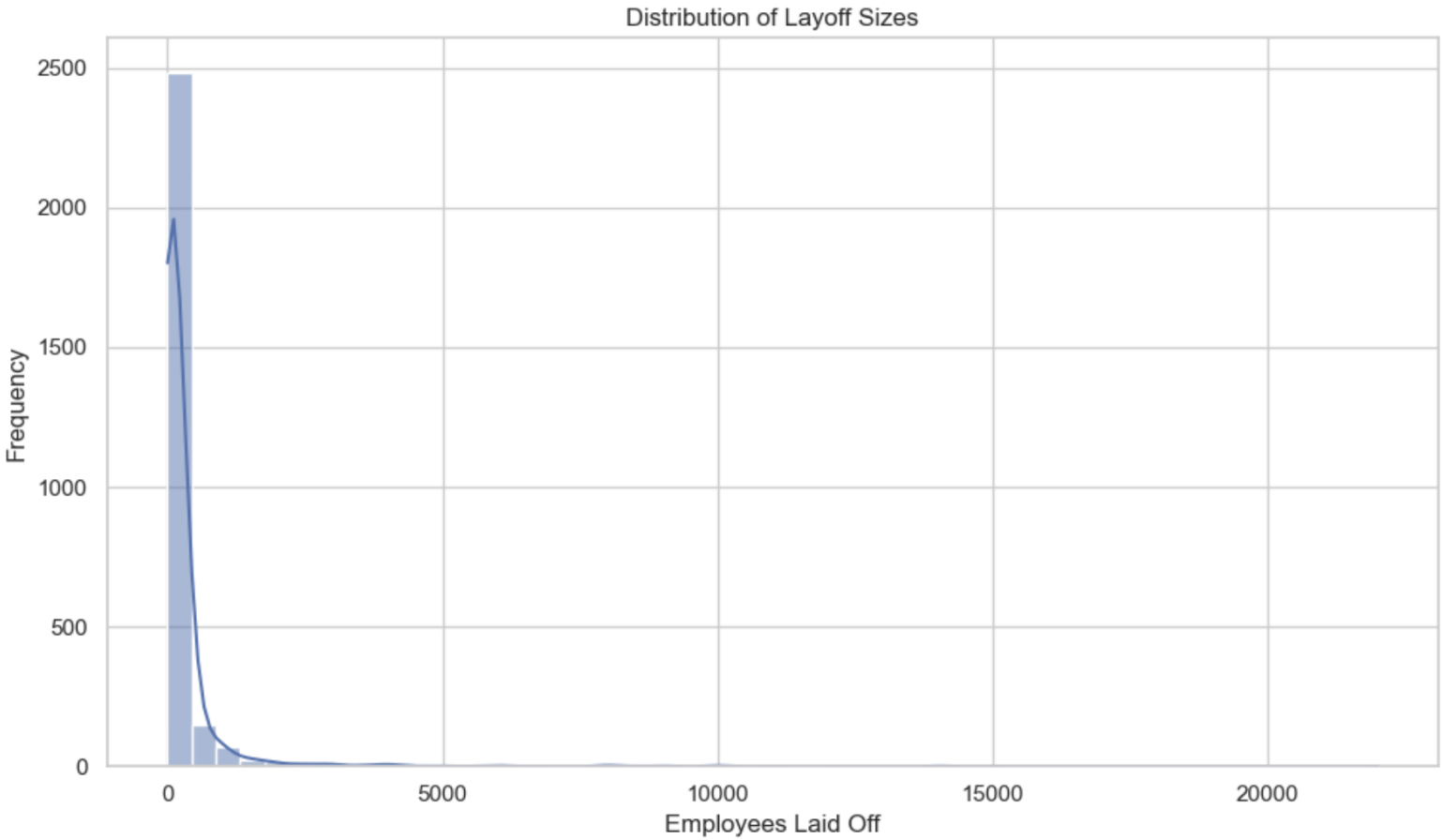
- Post-IPO companies drove total layoff volume
- Nearly 500,000 employees laid off post-IPO
- Early-stage firms faced higher % workforce loss
- Employee risk varies by company maturity

VISIBILITY BIAS METRIC

	industry	average_layoff_size	reported_events	visibility_index
20	Other	478.646409	290	138807.458564
12	Hardware	1827.588235	75	137069.117647
24	Retail	387.767544	341	132228.732456
3	Consumer	451.956522	275	124288.043478
28	Transportation	363.777778	264	96037.333333
8	Finance	160.190476	505	80896.190476
10	Food	323.636943	244	78967.414013
13	Healthcare	188.854369	326	61566.524272
14	Infrastructure	520.875000	77	40107.375000
29	Travel	324.583333	107	34730.416667

- Public data reflects media visibility
- Large firms dominate layoff narratives
- Smaller firms and sectors underrepresented
- Risk of misinterpretation without context

DISTRIBUTION OF LAYOFFS



- Layoff sizes are highly right-skewed
- Most events are relatively small
- Few extreme events drive totals and averages
- Medians and distributions matter

KEY TAKEAWAYS

- Layoffs in 2022 were **clustered**, not gradual
- Industry impact varied by **frequency and magnitude**
- Size alone does not reflect **severity or employee risk**
- Public layoff data reflects **visibility, not completeness**
- Ethical context is essential in people analytics